

ZoneGuard AI

90-Second Demo & Pitch Presentation Script — National Multi-Hazard Red-Zone Mapping & Pre-Disaster Relocation Intelligence

Target Duration: 90 Seconds (1.5 Mins)

Word Count: ~215 Words

Pace: ~140 WPM (Clear & Authoritative)

1. [0:00 – 0:15] The Hook & The Problem

■ ■ Visual Action: Open Landing Page dashboard; highlight live red-zone casualty counters and high-risk alerts.

"Every monsoon, vulnerable communities across fragile terrains like the Western Ghats face catastrophic landslides and land subsidence with zero early warning. Traditional disaster management only reacts **after** tragedy strikes. What if we could detect sub-surface land movement months in advance and relocate people **before** a disaster occurs?"

2. [0:15 – 0:40] Satellite Radar Fusion & Real-Time Monitoring

■ ■ Visual Action: Switch to GIS Command Center, navigate into Tamil Nadu red zones (e.g. Coonoor or Courtallam), and show multi-satellite radar HUD.

"Introducing **ZoneGuard AI** — a National Multi-Hazard Red-Zone Mapping and Pre-Disaster Relocation Intelligence System.

We continuously fuse Earth Observation satellites: **Copernicus Sentinel-1 SAR** detects microscopic ground creep at millimeter precision using PSInSAR interferometry, combined with **ISRO Cartosat DEM** for slope shear and **Landsat multispectral data** for soil saturation."

3. [0:40 – 1:05] TreeSHAP Explainable AI & Smart Relocation Engine

■ ■ Visual Action: Click 'Explain AI' (TreeSHAP modal), then click the green [Match Relocation Site →] button. Watch the map route corridor and haven appear with celebration confetti.

"With **TreeSHAP Explainable AI**, disaster officers don't just see an arbitrary risk score — they see exactly **why** a zone is failing.

With one click, our **Analytical Hierarchy Process (AHP)** engine pairs the compromised community with the safest relocation haven. It dynamically verifies carrying capacity, calculates evacuation corridors, and charts the safest transit route."

4. [1:05 – 1:30] Ground Ops, Field Verification & Closing Impact

■ ■ Visual Action: Toggle to Deformation Explorer showing dynamic PSInSAR displacement curves, then click 'Decision PDF' to demonstrate auditable export.

"From district collectors generating legally auditable relocation directives to ground officers on the front lines, ZoneGuard AI shifts disaster response from reactive relief to proactive protection.

ZoneGuard AI — Mapping invisible hazards, safeguarding human lives."

■ Presenter Quick-Reference Timeline

Timestamp	Phase	Live Screen Action	Key Talking Point
0:00 – 0:15	The Problem	Landing Page & Red-Zone Counters	Shift from reactive relief to pre-disaster prediction
0:15 – 0:40	Satellite Pipeline	GIS Command Center + Live Radar	Sentinel-1 PSInSAR mm creep + ISRO Cartosat DEM
0:40 – 1:05	Core AI Demo	TreeSHAP → Match Relocation Site	Explainable drivers & automated AHP haven routing
1:05 – 1:30	Impact & Close	InSAR Curve & Decision PDF Modal	District legal directives & saving human lives